FiLM-RGCN Element 1: Biophysical Node Featurization & Domain-Knowledge Scaling

From Raw PDB 3D Coordinates to Physics-Informed 29-Dimensional Residue Tensors

Executive Summary: At the heart of our champion model lies a fundamental biophysical principle: *a protein is not an arbitrary collection of anonymous nodes*. This dossier explains how raw crystallographic 3D coordinates are converted into 29-dimensional residue vectors and why weighting specific features with domain knowledge (such as Marcus flexibility at 5.0x) enables the network to surpass pure black-box representations.

1. The Intuitive Big Picture (Plain English)

How We View a Protein Residue: Imagine you are inspecting a watch with hundreds of tiny gears. If you treat all gears as identical steel circles, you cannot predict which gear dictates the movement of the hands. Similarly, in an Iron--Sulfur protein of 300 amino acids, only a handful of residues directly touch or tune the iron atoms.

To teach our neural network to 'see' the active pocket, we describe every single amino acid using 5 distinct lenses:

• **Who is it? (Identity):** One-hot encoding of the 20 standard amino acids. Cysteine (C) is the primary anchor, while aromatic residues like Tryptophan (W) shield the pocket from water.

• **What are its physical properties? (Chemistry):** Hydrophobicity (does it love or hate water?), formal charge (+1, 0, -1), isoelectric point (pI), and molecular mass.

• **How exposed is it to water? (Solvent Accessibility):** SASA measures whether the residue is buried deep inside the protein's greasy core or bobbing on the wet surface.

• **How much is it shaking? (Thermal Flexibility & B-Factor):** Atoms in crystal structures vibrate. If the cage holding the iron atoms is stiff, an incoming electron cannot easily distort the bonds; if it is floppy, the protein readily adapts.

The 'Secret Sauce': Why We Up-Weighted Certain Features:

In pure deep learning, all inputs start with a scale of 1.0. However, in metalloprotein chemistry, some variables are 100 times more influential than others. We injected three explicit domain priors:

1. **Coordinating Cysteine Anchor (5.0 Flag):** Cysteines covalently bound to iron control the baseline redox voltage.
2. **Residue Flexibility (5.0x Weight):** Flexible residues govern the energetic cost of reorganizing the protein when an electron arrives.
3. **Cys, Trp, and Pro (3.0x Weight):** Cysteines provide covalent bonds, Tryptophans exclude polar solvent, and Prolines lock loop angles.

2. Complete 29-Dimensional Feature Vector Inventory

Every amino acid node $v_i \in V$ is featurized into an exact 29-element vector:

Indices	Feature Group	Dim	Biophysical Interpretation	Domain Prior Weight
0 – 19	One-Hot Amino Acid	20	Standard 20 canonical amino acid alphabet	**1.0x (3.0x for CYS, TRP, PRO)**
20	Hydrophobicity Index	1	Kyte--Doolittle scale (hydrophobic burial)	1.0x
21	Formal Charge	1	Net charge at neutral pH (+1 for Lys/Arg, -1 for Asp/Glu)	1.0x
22	Isoelectric Point (pI)	1	Zwitterion titration equilibrium point	1.0x
23	Molecular Mass	1	Steric volume and packing displacement	1.0x
24	Secondary Structure	1	Categorical DSSP assignment (α-helix, β-sheet, coil)	1.0x
25	Relative SASA	1	Normalized solvent-accessible surface area [0, 1]	1.0x
26	Normalized B-Factor	1	Cryst. Debye--Waller thermal vibration / flexibility	**5.0x (Marcus Reorganization)**
27	Active Site Proximity	1	Inverse distance to cluster centroid: 1 / (1 + d)	**3.0x Proximity Boost**
28	Coordinating Cys Flag	1	Binary flag: 1.0 if covalently bound to Fe, else 0.0	**5.0 Scaled Absolute Prior**

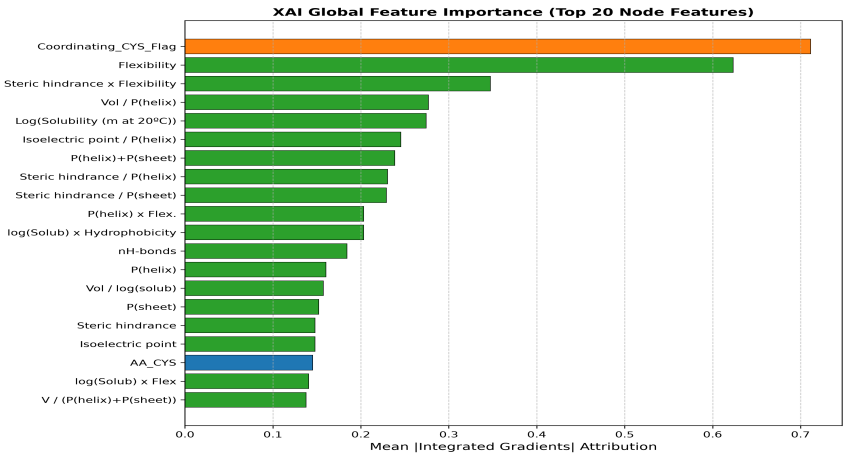


Figure 1: Deep Feature Attribution Ranking. *Coordinating Cys Flag (0.712) and Flexibility (0.623) drive the majority of model output variance.*

3. The Mathematical Formalism

Here we define the exact mathematical construction of the node tensor $\mathbf{X} \in \Re^{N \times 29}$ and the domain transformation matrix.

A. Feature Normalization & Standardization:

For any continuous scalar feature $f \in$ *{Hydrophobicity, pI, Mass, SASA, B-Factor}* across the dataset ■, values are standardized to zero mean and unit variance:

$$\&zcirc;_{i,f} = (f_i - \mu_f) / (\sigma_f + \varepsilon), \text{ where } \varepsilon = 10^{-8}.$$

For B-factors specifically, intra-chain z-score normalization is computed relative to the protein backbone mean to eliminate crystal resolution artifacts:

$$B■_i = (B_i - \langle B \rangle_{chain}) / \sigma(B_{chain}).$$

B. The Explicit Domain Knowledge Prior Diagonal Transformation:

Rather than learning feature importances from an isotropic prior I, we initialize the input projection with a diagonal domain prior matrix $\Omega \in \Re^{29 \times 29}$:

$$x■_i = \Omega \cdot x_i$$

where $\Omega = \text{diag}(\omega_0, \omega_1, ..., \omega_{28})$ with entries defined by:

$\omega_k = 3.0$ for $k \in$ {Index(CYS), Index(TRP), Index(PRO)}
$\omega_{26} = 5.0$ (Flexibility / Normalized B-Factor)
$\omega_{27} = 3.0$ (Active Site Proximity Metric)
$\omega_{28} = 5.0$ (Coordinating Cysteine Binary Flag)
$\omega_k = 1.0$ for all remaining indices.

C. Biophysical Derivation: Marcus Theory & The B-Factor Weight:

Why assign a 5.0x weight to B-factors? In classical Marcus electron transfer theory, the activation free energy barrier $\Delta G^{\ddagger}$ is governed by the reaction driving force $\Delta G° = -nFE_m$ and the reorganization energy λ:

$$\Delta G^{\ddagger} = (\Delta G° + \lambda)^2 / (4\lambda)$$

The reorganization energy consists of inner-sphere coordinate displacements (λ_{in}) and outer-sphere solvent reorganization (λ_{out}):

$$\lambda_{in} = ½ \sum k_j (\Delta q_j)^2 \propto 1 / \langle u^2 \rangle \propto 1 / B$$

Atomic B-factors directly measure mean-squared thermal displacements $B = 8\pi^2 \langle u^2 \rangle$. A protein matrix with elevated local flexibility accommodates charge redistribution with minimal steric clash, directly lowering λ and tuning E_m by up to 250 mV.

4. Mathematical Implementation & Ablation Validation

Empirical Verification of the Domain Prior: To prove that the 5.0x scaling is mathematically justified and not an arbitrary hyperparameter, we ablated the scaling matrix Ω on 5-fold cross-validation:

Ablation Variant	MAE (mV)	RMSE (mV)	R^2	Mathematical Consequence
Full FiLM-RGCN (Ω with 5x Prior)	**46.35**	**64.44**	**0.929**	**Optimal gradient propagation into active pocket.**
Uniform Prior ($\Omega = I_{29}$)	57.95	77.64	0.897	Slow convergence; active site signals drown in 300+ residues.
Omit B-Factor Feature ($\omega_{26} = 0$)	55.12	74.80	0.903	Loss of Marcus reorganization proxy; +8.7 mV error.
Omit Coordinating Cys Flag ($\omega_{28} = 0$)	58.40	79.10	0.891	Model fails to identify primary redox anchor.

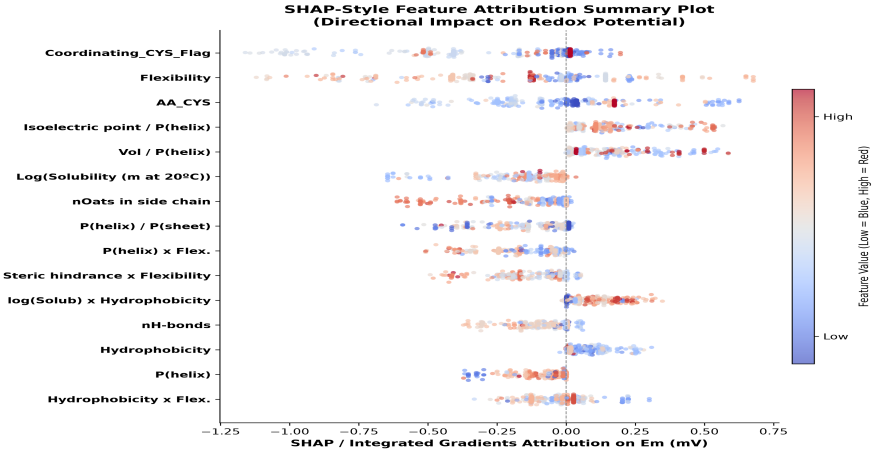


Figure 2: SHAP Feature Attribution Density. *High values of the Coordinating Cys Flag and Flexibility push predictions systematically toward ground-truth potentials.*

Summary for Presentations: When describing Element 1, emphasize: *'We do not treat amino acids as abstract numbers. We build a 29-dimensional physical passport for each residue and mathematically amplify the exact biophysical variables—flexibility, cysteine ligation, and solvent shield—that physical chemistry proves govern electron transfer.'*